Chapter 1: Foundations of Machine Learning and RAG

Machine learning algorithms build mathematical models based on sample training data.

Supervised learning trains algorithms on labeled input and output data pairs.

Unsupervised learning identifies hidden patterns or data groupings without human supervision.

Retrieval-Augmented Generation (RAG) is an architectural approach that combines dense vector retrieval with generative language models to provide grounded, factual responses with source citations.

FAISS (Facebook AI Similarity Search) is an efficient vector indexing and similarity search library.

Chapter 2: Vector Embeddings and Transformer Architectures

Embeddings map high-dimensional words, tokens, or passages into continuous dense vector representations.

The all-MiniLM-L6-v2 sentence transformer model produces 384-dimensional dense semantic vectors.

Cosine similarity measures the cosine of the angle between two multi-dimensional vectors.

In RAG pipelines, chunks with highest cosine similarity are passed as context to the generative LLM.

Evaluation metrics like Faithfulness evaluate whether the response contains only supported statements.